

MICHAEL LOZADA-LONGOG

Biomedical Engineering / BMET Intern

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PROFESSIONAL SUMMARY

Biomedical Engineering undergraduate at Hawaii Pacific University seeking BMET, clinical engineering, field service, medical device, or healthcare technology internships on Oahu. Hands-on experience in hospital operations, equipment-management workflows, CAD/FEA, Python, embedded systems, healthcare IT, and technical documentation. Builds independent engineering portfolio projects that connect clinical needs with practical design and data tools.

EDUCATION

Hawaii Pacific University

Bachelor of Science, Biomedical Engineering | GPA: 3.3

Relevant Coursework: General Biology, Chemistry, Introduction to Programming (Python)

Expected May 2029

Honolulu, HI

EXPERIENCE

Unit Clerk | Kuakini Medical Center

Jun 2025 - Sep 2025

Honolulu, HI

- Supported clinical unit operations by managing patient records, charts, and order entry in a hospital environment.
- Coordinated communication among nursing staff, departments, and patients while maintaining accurate, HIPAA-compliant documentation.

Social Media Intern | Hawaii Pacific University

Jan 2026 - Present

Honolulu, HI

- Produce institutional content for university-wide events and manage Instagram, TikTok, and LinkedIn content reaching an audience of 3,500+.
- Analyze platform metrics and adjust content strategy based on engagement and performance data.

Peer Administrative Assistant / Mail Room Assistant | Hawaii Pacific University

May 2026 - Present

Honolulu, HI

- Provide front-line support, scheduling, records management, package tracking, accurate data entry, and responsive service to students, faculty, and staff.

SELECTED ENGINEERING PROJECTS

Project TITAN - Mechanical Knee Support Assembly | Fusion 360 | Jul 2026

- Designed a multi-component parametric assembly with side plates, frames, shaft, spacers, bearings, and hardware; produced an exploded view and complete manufacturing drawing package.
- Performed static-stress FEA and validated approximately 52.9 MPa maximum Von Mises stress, 2.377 minimum safety factor, and 0.004 in maximum displacement.
- Documented material selection, bill of materials, manufacturing strategy, design validation, and future improvements in an 11-page engineering report.

Project 01 - Kapolei BioForge Equipment Tracker | Excel / Netlify | Jul 2026

- Built a QR-based mock biomedical equipment management system for 15 assets with individual records, service notes, inspection schedules, printable QR labels, and mobile-accessible pages.
- Created an Excel operations dashboard and deployed the project through a public Netlify portfolio site.

Project 02 - BMET PM & Service Ticket System | Excel | Jul 2026

- Developed a mock CMMS-style workbook linking 20 service tickets, 15 equipment assets, and 12 parts/consumables across preventive and corrective maintenance workflows.
- Automated SLA status, downtime, root cause, parts cost, safety verification, return-to-service status, PM alerts, and inventory reorder reporting.

Project 03 - Medical Equipment Troubleshooting & Failure Analysis Simulator | Excel | Jul 2026

- Built an interactive simulator with 12 fictional equipment cases and seven scored decisions per case covering safety, diagnosis, root cause, corrective action, verification, and return-to-service documentation.
- Created an 18-item FMEA table with automatic Risk Priority Number calculations, a scorecard, failure-analysis log, and performance dashboard.

ADDITIONAL TECHNICAL PROJECTS

Integrated Clinical Systems Suite | *Python* | 2026

- Engineered a 24-module Python suite integrating diagnostic imaging physics, signal processing, fluid mechanics, nuclear medicine math, healthcare IT, machine learning, and a centralized terminal dashboard.
- Implemented CT, X-ray, ultrasound, pacemaker, ventilator, infusion pump, anesthesia, dialysis, defibrillator, PACS/DICOM, HIPAA anonymization, and medical-device security simulations with GitHub Actions CI/CD.

Postural Sway Fall Risk Analyzer | *Raspberry Pi / Fusion 360 / XGBoost* | May 2026

- Designed a wrist-worn Raspberry Pi Zero 2W and MPU-6050 enclosure in Fusion 360 and built an XGBoost dashboard producing a 0-100 fall-risk score from sway angle, velocity, and XYZ acceleration data.

Biomedical Data & Machine Learning Portfolio | *Python* | 2026

- Completed diabetes and heart-disease risk modeling, Pacific Islander genomics/pharmacogenomics analysis, and a 55,500-patient healthcare admissions study using Python, pandas, scikit-learn, XGBoost, Random Forest, Logistic Regression, and statistical testing.

ACTIVITIES & LEADERSHIP

NSBE Chapter Founder | *Hawaii Pacific University*

2026 - Present
Honolulu, HI

- Initiating HPU's first NSBE collegiate chapter and coordinating with Region VI leadership and Hawaii/SoCal representatives.

Founder | *Kapolei BioForge*

2026 - Present
Kapolei, HI

- Publish biomedical engineering builds, BMET workflow systems, CAD/FEA projects, medical equipment simulations, and biomedical AI research through a public portfolio and social platforms.

IEEE Student Branch Member | *Hawaii Pacific University*

2026 - Present
Honolulu, HI

- Participate in HPU IEEE activities connecting engineering students with technical resources and industry networks.

CERTIFICATIONS & MEMBERSHIPS

- Stanford x DeepLearning.AI Machine Learning Specialization - Andrew Ng (May 2026)
- BMET 101: CABT Prep Course - AAMI (Enrolled 2026)
- AAMI Student Member - Association for the Advancement of Medical Instrumentation (2026)
- IEEE Student Member and IEEE EMBS Student Member (2026)

SKILLS

BMET & Clinical Engineering: Preventive maintenance scheduling, service ticket workflows, asset management, CMMS-style documentation, downtime and SLA tracking, troubleshooting logic, FMEA, root-cause analysis, safety verification, return-to-service documentation

CAD & FEA: Fusion 360, SolidWorks, parametric modeling, assembly design, static-stress FEA, Von Mises stress, safety factor analysis, engineering drawings, design for manufacturing, 3D printing

Clinical & Healthcare IT: PACS/DICOM, HL7, HIPAA compliance, medical device simulation, patient telemetry, healthcare operations

Programming & Data: Python, Excel, XGBoost, Random Forest, Logistic Regression, scikit-learn, pandas, NumPy, Git, GitHub Actions, HTML/CSS/JavaScript

Embedded & DSP: Raspberry Pi Zero 2W, MPU-6050 IMU, I2C, GPIO, soldering, IIR notch filters, LMS adaptive filters, FFT, ECG/PPG/SpO2 modeling